A two-dimensional obstruction for the Banach projection angle

Run fa_banach_001, agent_lane_15

June 28, 2026

Abstract

Oppenheim defines a Banach-space analogue of the cosine of the angle between two projections and remarks that it is not known in general whether this quantity is bounded by one. We give an explicit pair of projections on $\mathbb{R}^2$ for which the definition gives $\sqrt{2}$. The same pair also shows that the corresponding angle between the complementary projections need not be well defined.

1 Source question

The source paper is Oppenheim's arXiv:1512.08188v2, "Vanishing of cohomology with coefficients in representations on Banach spaces of groups acting on Buildings" [1]. Definition 2.10 defines, for projections P_1, P_2 onto M_1, M_2, the quantity

$$\cos(\angle(P_1, P_2)) = \max\{\|P_1(P_2 - P_{1,2})\|, \|P_2(P_1 - P_{1,2})\|\},$$

provided there is a projection $P_{1,2}$ onto $M_1 \cap M_2$ with $P_{1,2}P_1 = P_{1,2}$ and $P_{1,2}P_2 = P_{1,2}$. Remark 2.11 says that it is unknown whether this "cosine" is always at most 1.

> *$P_{1,2}$ on $M_1 \cap M_2$ such that $P_{1,2}P_1 = P_{1,2}$ and $P_{1,2}P_2 = P_{1,2}$ and define*
>
> $$\cos(\angle(P_1, P_2)) = \max\{\|P_1(P_2 - P_{1,2})\|, \|P_2(P_1 - P_{1,2})\|\}.$$
>
> **Remark 2.11.** *In the above definition, we are actually defining the "cosine" of the angle. This is a little misleading, because we do not know if $\cos(\angle(P_1, P_2)) \leq 1$ holds in general (although this inequality holds in all the examples we can compute or bound).*

Remark 3.11 later notes an additional obstruction: for Banach-space projections, it is not clear whether the angle between complements is even defined.

In particular, $\cos(\angle(P_\tau, P_{\tau'})) \leq \varepsilon$.

Remark 3.11. *Variations of the above theorem were proven in the setting of Hilbert spaces in [8], [9] and [13]. However, all these proofs use the fact that in a Hilbert space the following equality holds for any two subspaces* U, V*:* $\angle(V, U) = \angle(V^\perp, U^\perp)$*, where the angle here is the Friedrichs angle. In our setting, we do not know if such equality holds, namely if* $\cos(\angle(P_\tau, P_{\tau'})) = \cos(\angle(I - P_\tau, I - P_{\tau'}))$ *(we don't even know if* $\cos(\angle(I - P_\tau, I - P_{\tau'}))$ *is well defined). This limitation required us to give a more direct proof using the idea of angle between several projections.*

2 Idea of the proof

The definition controls products of the chosen projections, not merely the angle between their ranges. If one projection is oblique, its norm can be larger than one even in a Hilbert space. Taking the two ranges to be distinct one-dimensional lines makes their intersection zero, so the required intersection projection is just the zero map. The source definition then reduces to the maximum of two product norms. One of these products is exactly the oblique projection, whose Euclidean norm is $\sqrt{2}$.

3 Counterexample

Theorem 1. *There exist a Banach space* X*, two bounded projections* P_1, P_2 *on* X*, and an admissible projection* $P_{1,2}$ *in the sense of Definition 2.10 of [1], such that*

$$\cos(\angle(P_1, P_2)) > 1.$$

Indeed this value can already be $\sqrt{2}$ *on* $X = \mathbb{R}^2$ *with the Euclidean norm.*

Proof. Let $X = \mathbb{R}^2$ with its Euclidean norm and define

$$P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

Equivalently, $P_1(x, y) = (x, 0)$ and $P_2(x, y) = (x, x)$. Both maps are projections, since $P_1^2 = P_1$ and $P_2^2 = P_2$. Their ranges are

$$M_1 = \operatorname{span}(1, 0), \qquad M_2 = \operatorname{span}(1, 1),$$

and hence $M_1 \cap M_2 = \{0\}$. Thus $P_{1,2} = 0$ is a projection onto the intersection and satisfies the two required identities

$$P_{1,2}P_1 = P_{1,2}, \qquad P_{1,2}P_2 = P_{1,2}.$$

The definition therefore gives

$$\cos(\angle(P_1, P_2)) = \max\{\|P_1P_2\|, \|P_2P_1\|\}.$$

Now $P_1P_2 = P_1$, so $\|P_1P_2\| = 1$, while $P_2P_1 = P_2$. The map P_2 is the rank-one operator $e_1^* \otimes (1, 1)$, and its Euclidean operator norm is

$$\|P_2\| = \|e_1^*\| \, \|(1, 1)\| = \sqrt{2}.$$

Consequently

$$\cos(\angle(P_1, P_2)) = \sqrt{2} > 1.$$

□

Proposition 1. *For the same P_1, P_2, the source definition of $\cos(\angle(I-P_1, I-P_2))$ is not applicable: there is no projection onto $\operatorname{im}(I-P_1) \cap \operatorname{im}(I-P_2)$ satisfying the two compatibility identities required in Definition 2.10.*

Proof. Put

$$Q_1 = I - P_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Q_2 = I - P_2 = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}.$$

Both Q_1 and Q_2 are projections and both have range $\operatorname{span}(0,1)$. Hence the relevant intersection is again $\operatorname{span}(0,1)$.

Every projection R onto this line has the form

$$R_c = \begin{pmatrix} 0 & 0 \\ c & 1 \end{pmatrix}$$

for some scalar c. If $RQ_1 = R$, then applying both sides to e_1 gives $0 = ce_2$, so $c = 0$. Thus $R = Q_1$. But then

$$RQ_2 = Q_1 Q_2 = Q_2 \neq Q_1 = R.$$

Therefore no such R can satisfy both $RQ_1 = R$ and $RQ_2 = R$, so the angle between the complementary projections is not well defined under the source definition. □

4 Verification notes

The example is finite dimensional, and all identities are exact matrix identities. The included script `code/rank_two_projection_angle_check.py` verifies the idempotence, the product norms, and the obstruction to the complementary intersection projection. The value $\sqrt{2}$ is exact; the numerical output is only a sanity check.

5 Novelty and limitations

Local run indexes were searched for arXiv:1512.08188, the title, and the core keywords "angle between projections", "Friedrichs angle", and "averaged projections"; no exact prior packet or attempt was found. Bounded web searches on June 28, 2026 for exact phrases around $\cos(\angle(P_1, P_2)) \leq 1$, Oppenheim's projection-angle cosine, and arXiv:1512.08188 plus "angle between projections" found no separate literature answer beyond the source arXiv record.

The example does not challenge the theorems in [1], which assume small numerical bounds on the defined cosine when they use it. It only settles the two general definitional questions raised in Remarks 2.11 and 3.11: the quantity need not be bounded by one, and complement angles need not be defined for arbitrary bounded projections.

6 Human-review recommendation

The key point for review is whether Definition 2.10 is intended to allow arbitrary bounded projections, including non-orthogonal projections on a Hilbert space. The surrounding text says "projection" in the Banach-space sense $P^2 = P$, so the example appears to be a literal counterexample to the general question in Remark 2.11.

References

[1] I. Oppenheim, *Vanishing of cohomology with coefficients in representations on Banach spaces of groups acting on Buildings*, arXiv:1512.08188v2, 2016. `https://arxiv.org/abs/1512.08188`.